

# Method of Handling Traffic — Crew Instructions

## Project Information

- **Speed limit:** 40 mph
- **Road type:** Urban (low-speed)
- **Closure type:** Flagger Alternating Traffic — 2-Lane Undivided
- **Work zone length:** 3,000 ft
- **Jurisdiction:** Littleton
- **Generated:** 2026-08-22

## Required Equipment

- 20× Construction/Warning Sign (see sign list below)
  - 2× G20-1 ROAD CONSTRUCTION (NEXT 3000 FT)
  - 2× G20-2 END ROAD WORK
  - 2× G20-5P CONSTRUCTION ZONE plaque
  - 4× R9-9 SIDEWALK CLOSED
  - 2× S1-1 SCHOOL
  - 2× W20-1 ROAD WORK AHEAD
  - 2× W20-4 ONE LANE ROAD AHEAD
  - 2× W20-7a AFAD AHEAD
  - 2× W3-4 BE PREPARED TO STOP
- 6× Channelizing Drum (36-inch)
- 43× Traffic Cone (28-inch)
- 6× Type C Steady-Burn Warning Light
- 4× Type III Barricade
- 1× Portable Light Plant
- 2× Temporary Traffic Signal
- 1× Pilot car with G20-4 "PILOT CAR/FOLLOW ME" sign mounted on the rear of the vehicle (S-630-1 Sheet 26 — vehicle-mounted, not a roadside placement)

## Sign Placement Schedule

| Sign Code | Description              | Distance from Taper     |
|-----------|--------------------------|-------------------------|
| S1-1      | SCHOOL                   | 1,000 ft upstream       |
| W20-1     | ROAD WORK AHEAD          | 500 ft upstream         |
| W20-4     | ONE LANE ROAD AHEAD      | 400 ft upstream         |
| W3-4      | BE PREPARED TO STOP      | 300 ft upstream         |
| W20-7a    | AFAD AHEAD               | 200 ft upstream         |
| G20-5P    | CONSTRUCTION ZONE plaque | Within work zone        |
| G20-2     | END ROAD WORK            | Downstream of work zone |

## Setup Procedure

**Important: Set up from downstream to upstream (against traffic flow). This ensures you are always working behind protection.**

1. Identify both AFAD station positions (do not staff yet — stations are staffed after all devices are in place, step 7): Station #1 at station 3,505 ft (approach side, 100 ft upstream of the taper start, closed-lane side); Station #2 at station -300 ft (opposing approach, 300 ft past the work-area end per CDOT S-630-1 Case 17).
2. Place the END ROAD WORK sign (G20-2) at the downstream termination (station -150 ft), right side of road 14 ft from centerline. Place the opposing-direction G20-2 past the upstream end of the work zone (left side).
3. Place 4 traffic cones in the downstream taper, between station -50 ft (right lane edge) and station 0 (centerline) at 12 ft spacing — this taper returns right-direction traffic from the centerline back to the right lane edge past the work area (MUTCD §6B.08 ¶12; S-630-1 Case 17 annotates the run 100' MAX.). Then place 39 traffic cones along the centerline boundary between the open opposing lane and the closed work lane at 79 ft spacing, starting at station 0 (downstream end of work zone) and working upstream to station 3,000 ft.
4. Place G20-5P WORK ZONE plaque(s) within the work zone at approximately 1,500 ft intervals.
5. Place 6 channelizing drums in the one-lane two-way taper, starting at station 3,305 ft and working upstream to station 3,405 ft. Space drums at 20 ft intervals (approximately 20 ft per MUTCD §6B.08 — this taper stages stopped traffic behind the flagger; it must be short enough to not be mistaken for a transition). Drums transition from the right lane edge (10 ft) to the centerline (0 ft).
6. Place advance warning signs upstream of the taper: Repeat the series for the opposing approach (left side, negative stations — S-630-1 Case 17: "SIGN SEQUENCE IS THE SAME FOR OPPOSITE DIRECTION").
  - W20-7a at 200 ft upstream of taper start (station 3,605 ft), right side.
  - W3-4 at 300 ft upstream of taper start (station 3,705 ft), right side.
  - W20-4 at 400 ft upstream of taper start (station 3,805 ft), right side.
  - W20-1 at 500 ft upstream of taper start (station 3,905 ft), right side.
  - S1-1 at 1,000 ft upstream of taper start (station 4,405 ft), right side.
7. Staff both AFAD stations (position and activate the AFADs; the operator must maintain sight of both devices), with station flood lighting per S-630-1 Sheet 2 Note 22. Verify each station is visible to approaching traffic from the full advance-warning distance and has an identified escape path. Begin the alternating one-way operation.

## Takedown Procedure

**Important: The one-lane operation ends before its warning signs come down. Flaggers control traffic until the roadway is fully restored to two-way operation.**

1. Complete work and clear all personnel and equipment from the work area.
2. Under continued flagger control, remove the work-zone cones and the taper drums (restoring the closed lane).
3. Cease the alternating one-way operation and release both AFAD stations.
4. Remove advance warning signs on both approaches: S1-1 first, then W20-1, then W20-4, then W3-4, then W20-7a.
5. Remove G20-5P plaques and the G20-1/G20-2 BEGIN/END ROAD WORK signs.

## Safety Notes

- All personnel in the work zone must wear ANSI Class 2 (daytime) or Class 3 (nighttime) high-visibility apparel.
- Nighttime operation: Ensure all devices meet retroreflectivity requirements. Flagger stations require 500W quartz illumination at minimum 8 ft height per CDOT S-630-1 Sheet 2 General Note 22.
- Maintain 305 ft longitudinal buffer space between the taper and the work area at all times.
- Maximum speed reduction per sign installation: 15 mph (CDOT S-630-1 Sheet 2 General Note 3).

- This plan was generated by MHT Tool. A certified Traffic Control Supervisor (TCS) must review, approve, and sign before field implementation.
- Reference: CDOT S-630-1, MUTCD 11th Edition Part 6, Colorado Supplement (effective January 18, 2026).

## Pedestrian and Bicycle Accommodation

- A pedestrian facility is affected by this closure. 4 R9-9 SIDEWALK CLOSED signs are placed at the closure (MUTCD §6G.10 — the plain R9-9 legend; no alternate-side assertion, which belongs to R9-10, and no detour-routing signs, which require sidewalk geometry this plan does not carry). Sidewalk closure points are barricaded (MUTCD §6C.02). Pedestrians must not be diverted into an open travel lane.
- **Littleton LEDS §131** — "An accessible pedestrian and bicycle route, per MUTCD standards, shall be maintained adjacent to the work area, at all times. Pedestrians shall not be diverted onto the roadway ... unless a detour route has been previously approved on the Traffic Control Plan."
- The TCS verifies pedestrian and bicycle accommodation on site; where a facility is maintained, it must remain accessible and detectable per MUTCD standards.

## Emergency Access

- The roadway is reduced to one alternating lane; access through the closure is controlled at the flagger stations.
- The Traffic Control Supervisor completes and confirms the emergency access plan for this site and coordinates notification of police, fire, and emergency medical responder agencies prior to work (TCS duties per CDOT Standard Specifications 630.11).
- An emergency access plan is a required minimum element of the approved Method of Handling Traffic (CDOT 630.10(a)); this section is draft input for it, not the approved plan.

## Site-Specific Notes

- **adjacent\_intersection** — No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure. (*MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)*)
- **adjacent\_interchange** — No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure. (*MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)*)
- **pedestrian\_facility** — Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone. (*MUTCD §6C.02 — pedestrian considerations in work zones*)
- **school\_zone** — Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 4,405 ft, both sides). (*MUTCD §7B.08 — school zone signing in proximity to work zones*)

## Night Operation Notes

- **night\_taper\_lighting** — Added 6 Type C steady-burn warning lights on taper drums for nighttime visibility. (*MUTCD §6L.07 — warning lights on channelizing devices in nighttime work zones*)
- **night\_work\_area\_lighting** — Added 1 portable light plant at the start of the work zone (25 ft past the buffer) to illuminate the work area. (*CDOT M&S §630.05.1 — illumination requirement for nighttime work zones*)
- **night\_retroreflective** — All devices, signs, and barricades must be retroreflective. Verify warehouse inventory before deployment. Inspect retroreflectivity at shift start. (*MUTCD §6F.01 + §6K.01 — retroreflectivity requirements for nighttime visibility*)